

SFL BRE Portal — System Architecture & Workflow Report

Organization: Satin Finserv Limited (SFL)

System: AI-Powered Business Rule Engine (BRE) & Credit Risk Platform

Version: v3.4 Production Stack

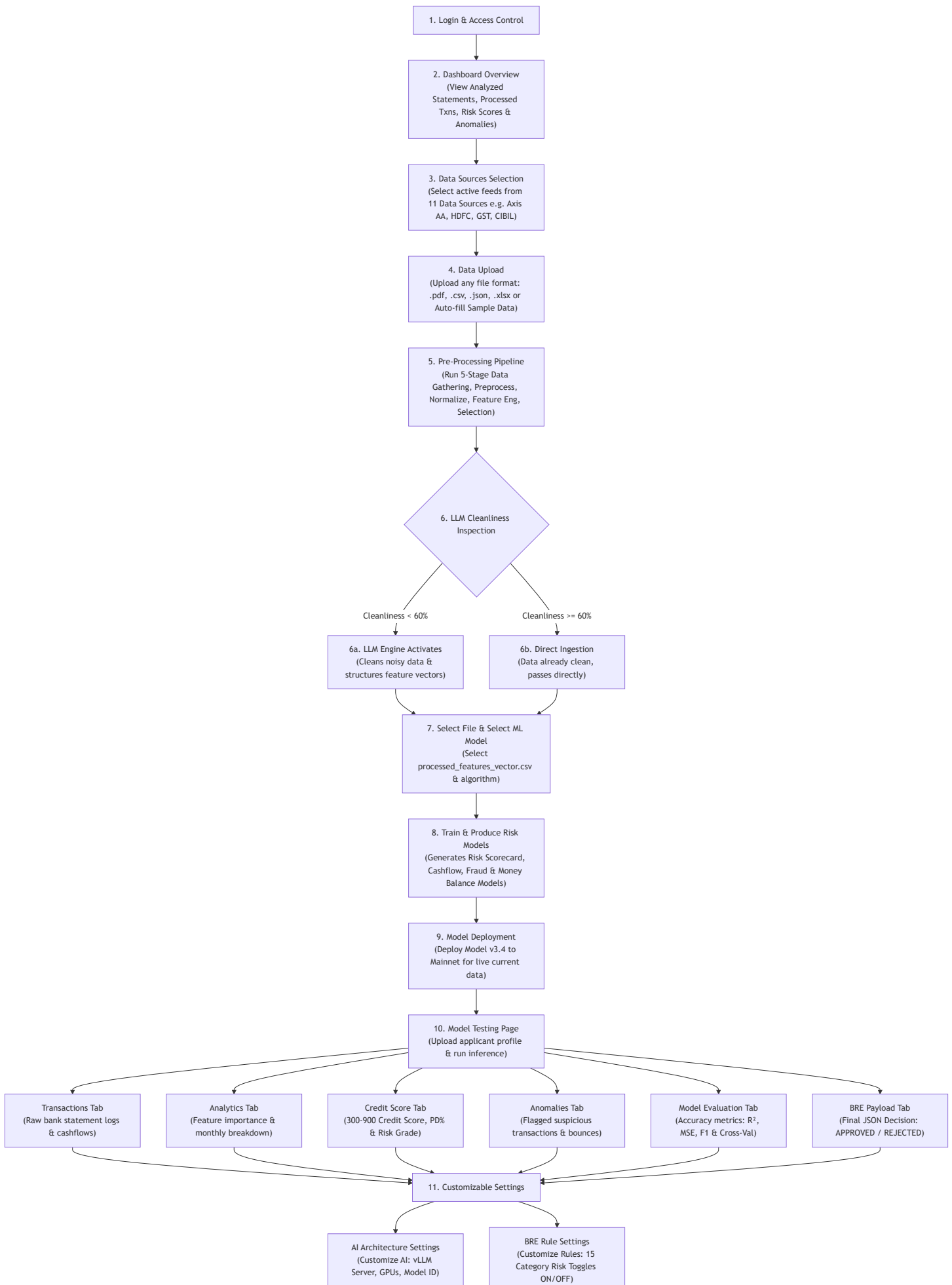
1. Project Objective & System Overview

The **Satin Finserv Limited (SFL) BRE Portal** is an enterprise-grade automated credit underwriting and risk decisioning platform. It automates loan evaluation by aggregating multi-source financial feeds, cleansing noisy raw data using local LLM engines, executing 5-stage feature engineering, training machine learning risk scorecards, enforcing customizable credit policy rules, and generating instant credit decision payloads.

Key Capabilities (What the Platform Does):

- **Multi-Feed Ingestion:** Aggregates 11 data vectors (Axis Bank AA, HDFC, SBI, ICICI, GST Returns, CIBIL Bureau, CERSAI, UPI Velocity).
- **Automated LLM Data Cleansing:** Evaluates raw data noise; if cleanliness is below 60% (noise > 40%), the LLM Data Cleaning Engine automatically activates to sanitize fields and engineer features.
- **Machine Learning Risk Modeling:** Trains 4 specialized sub-models (Credit Risk Scorecard, Cashflow Predictor, Fraud Detector, Liquidity Model) using algorithms like Gradient Boosting and Random Forest.
- **Real-Time Underwriting & Risk Scoring:** Calculates 300–900 credit scores, Probability of Default (PD%), risk grades (LOW, MEDIUM, HIGH), and flags suspicious transactions.
- **Customizable BRE Policy Rules:** Allows credit managers to toggle individual risk rules ON/OFF across 15 Underwriting Rule Categories.

2. End-to-End System Workflow Diagram



3. Platform Modules Overview

Module Name	Purpose & Primary Function
Overview Dashboard	Provides senior management with live portfolio KPIs, bar charts, donut charts, and bank statement ingestion feeds.
Data Sources	Enables selecting active financial feeds from 11 data vectors and adding custom feeds.
Model Hub	Handles raw file upload, pre-processing, LLM cleanliness filtering, ML model training, and version deployment.
Model Testing	Conducts live applicant risk scoring across 6 tabs (Transactions, Analytics, Credit Score, Anomalies, Model Evaluation, BRE Payload).
AI Architecture	Configures self-hosted vLLM engine settings (vLLM Endpoint, Model ID, GPU allocation, memory utilization).
Settings	Configures credit underwriting policies by toggling individual rules ON/OFF across 15 Risk Categories.